



IILM UNIVERSITY

GREATER NOIDA, UTTAR PRADESH

INDUSTRY INTERNSHIP REPORT

Full Stack Web Development

An 8-Week Professional Internship Presentation Report

INTERNSHIP DURATION

01 June 2026 – 31 July 2026 (8 Weeks)

STUDENT CREDENTIALS

Md Asrar

Full Stack Developer Intern

EMPLOYEE ID

EX321151

CERTIFICATE ID

ESSPT321151

WORK MODE

Remote / WFH

ROLL NUMBER

2410031686

PROGRAM & BATCH

B.Tech (CSE) • 3CSE19

HOST ENTERPRISE

Earthix Software Solutions Pvt. Ltd.

Corporate Verification: www.earthix.in | Authorized HR

Candidate & Program Overview

PROFILE DOSSIER



Candidate Profile

Verified intern documentation

Intern Name: Md Asrar

Employee ID: EX321151

University: IILM University, Greater Noida

Assigned Role: Full Stack Developer Intern

Contact Phone: +91 8340452870

Internship Mode: Work From Home (WFH)

Attendance Criteria: 100% Satisfactory Completed



Program Specification

Corporate internship details

Host Company: Earthix Software Solutions Pvt. Ltd.

Technical Domain: Full Stack Web Development

Core Architecture: MERN / MEAN Stack Systems

Tenure Duration: 2 Months (01 June – 31 July 2026)

Certificate ID: ESSPT321151

Issuance Date: 01 August 2026

Verification Portal: www.earthix.in

Organization Profile: Earthix Software Solutions

HOST ENTERPRISE

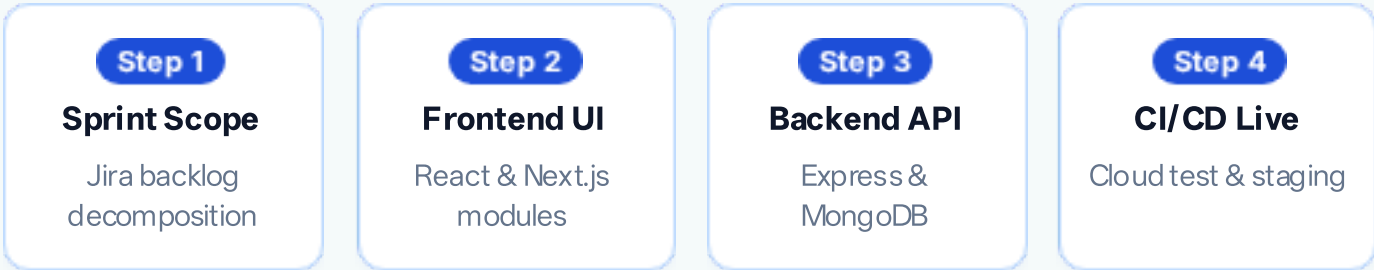


Corporate Mission & Focus

Enterprise software engineering

Earthix Software Solutions Pvt. Ltd. delivers robust cloud platforms, web applications, and scalable digital systems using agile sprints and rigorous code audits.

🔧 ENGINEERING DELIVERY WORKFLOW FLOWCHART



Core Engineering Pillars

Technical competencies & delivery

</>	Full-Stack Engineering: Modern JS Ecosystem
📄	Scalable Backends: Node.js & Express APIs
🗄️	Data Systems: MongoDB & PostgreSQL
🛡️	Security Standards: JWT & OAuth2 Protocols
☁️	Cloud Pipelines: Docker & Automated CI/CD
🌐	Official Portal: www.earthix.in

Problem Statement & Industry Context

STRATEGIC SCOPE



The Engineering Challenge

Academia vs. Real-World Web Apps

Academic projects often lack concurrency, indexing, and authorization handling.
Production web apps demand high performance:



Latency & Rendering: Heavy DOM manipulation causes UI freezes on mobile screens.



Security Risks: Vulnerabilities due to unvalidated inputs and absent JWT token checks.



Database Bottlenecks: Unindexed queries cause high response times under load.



Internship Solution Scope

Delivered at Earthix Software Solutions

Constructed an enterprise-grade web application architecture bridging modern frontends with secure, scalable backends:



Modular React Components: High Lighthouse speed & responsive UI across all viewports.

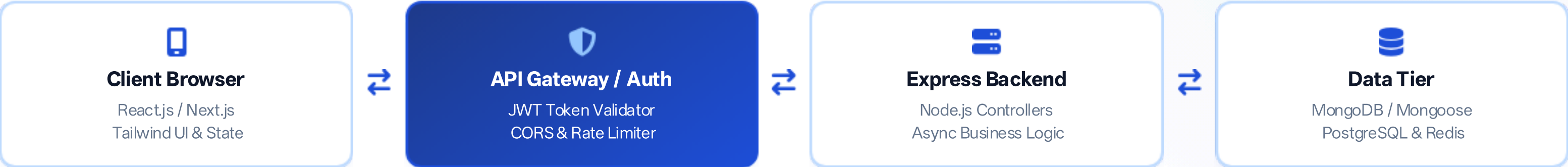
Stateless REST & Auth: Node/Express with bcrypt password hashing and JWT middleware.



Optimized Database: Mongoose schemas, compound indexes, and ACID transactions.

Full Stack Web Architecture & Data Flow

SYSTEM FLOWCHART



1. Presentation Flow

Client sends asynchronous JSON REST calls via Axios. Manages optimistic UI updates, responsive breakpoints, and client-side form sanitization.

2. Application Logic

Express server verifies incoming bearer tokens, validates payloads against schemas, applies business algorithms, and handles structured errors.

3. Persistence & Caching

Mongoose ODM coordinates query indexing, aggregation pipelines, and atomic updates. Persistent storage safeguards multi-tenant records.

Core Internship Objectives

MILESTONE GOALS



Frontend Architecture

Develop responsive, mobile-first web user interfaces using React.js, Next.js, and modern CSS frameworks, adhering to WCAG usability benchmarks.



Backend & REST APIs

Architect robust, stateless backend APIs with Node.js and Express.js, featuring middleware validation, logging, and error handling.



Database Schema Design

Implement optimized document schemas in MongoDB and relational models in PostgreSQL, ensuring data integrity and query efficiency.



Git & Full Deployment

Maintain industry-standard Git workflow (feature branching, PR reviews) and participate in live cloud deployment and staging tests.

Full Stack Technologies & Toolchain

TECHNICAL STACK



Frontend Tier

HTML5, CSS3, JavaScript (ES6+)

React.js, Next.js Framework
Tailwind CSS, Component UI Libraries
Redux Toolkit & Context API



Backend Tier

Node.js Runtime & Express.js

RESTful API Design Standards
JWT (JSON Web Tokens) Auth
Async Middleware & Error Handling



Database Systems

MongoDB (Mongoose ODM)

MySQL & PostgreSQL Relational DBs
Index Optimization & Aggregations
Data Validation & Migration



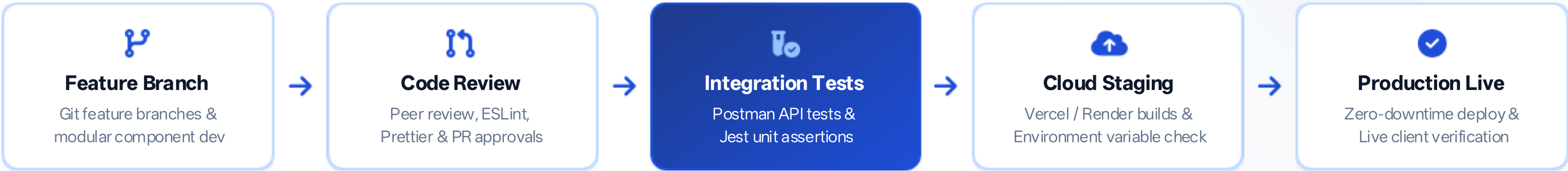
Tools & DevOps

Git & GitHub Collaboration

Postman API Client & Newman
Vercel, Render & AWS Cloud Hosting
Agile Scrum & Jira Workflows

Software Development & CI/CD Lifecycle

AGILE PIPELINE



Agile Collaboration at Earthix

Conducted daily stand-ups and sprint retrospectives. All features required modular task decomposition, standard semantic Git commits, and code review approvals by senior engineering leads.



Quality Assurance Benchmarks

Every endpoint was audited for injection vulnerabilities, strict header handling (Helmet.js), input sanitization, and responsive cross-browser testing across mobile, tablet, and desktop.

Internship Timeline: Weeks 1 to 4

PHASE 1 EXECUTION

Week 1

Onboarding & Tooling

Company orientation, Git repo setups, environment initialization, review of development guidelines and Earthix coding standards.

Week 2

Frontend Engineering

Developing responsive UI layouts with HTML5, modern CSS, Flexbox, CSS Grid, and reusable React.js functional components.

Week 3

Advanced React & State

Implementing client-side routing with React Router, Context API state management, form validation, and responsive mobile adaptations.

Week 4

Node & REST Foundations

Setting up Express servers, designing RESTful routing paradigms, implementing async handlers, and connecting with Postman for testing.

Internship Timeline: Weeks 5 to 8

PHASE 2 EXECUTION

Week 5

Database Modeling

Architecting MongoDB collections via Mongoose ODM, creating schemas, indexes, compound keys, and handling relational cross-references.

Week 6

Security & Authentication

Implementing bcrypt password encryption, JSON Web Tokens (JWT), role-based middleware, and protecting critical REST endpoints.

Week 7

Full Stack Integration

Connecting React frontend with Express backend via Axios, handling state synchronizations, loading loaders, and API error toasts.

Week 8

Testing & Live Deployment

Bug squashing, cross-browser compatibility checks, performance profiling, final documentation, and live hosting on cloud platforms.

Key Projects & Demonstrated Competencies

PRACTICAL OUTPUTS

✓ **Interactive Responsive Web Applications**

Engineered dynamic and modular user interfaces using React.js and Tailwind CSS, achieving high Lighthouse performance and mobile responsiveness across multiple device viewports.

✓ **Scalable REST API Endpoints with Node.js & Express**

Developed full CRUD backend services with centralized error logging, request payload validation, and clean async/await controller architecture.

✓ **Secure Authentication & Authorization Middleware**

Integrated token-based security pipelines using JWT and encrypted storage via bcrypt, restricting unauthorized data exposure across multi-tenant user tiers.

✓ **Optimized Database Schemas in MongoDB & PostgreSQL**

Designed and managed database operations with optimized indexing, reducing query retrieval latency and maintaining data integrity during continuous write operations.

✓ **Collaborative Agile Delivery & Version Control**

Utilized Git for structured commits, feature branching, code reviews with senior engineering mentors, and participating actively in weekly progress reviews.

Official Verification & Internship Metrics

CREDENTIAL MATRIX

Verification Metric	Official Record / Credential Details
Intern Name	Md Asrar
Employee ID	EX321151
Academic Institution	IILM University, Greater Noida, Uttar Pradesh
Internship Position	Full Stack Developer Intern
Host Organization	Earthix Software Solutions Pvt. Ltd. (CIN: Registered Private Limited)
Internship Period	01 June 2026 to 31 July 2026 (Duration: 2 Months / 8 Weeks)
Certificate Serial Number	ESSPT321151 (Issued: 01/08/2026)
Official Signatory	Director Md Sanaullah, Earthix Software Solutions Pvt. Ltd.

Internship Certificate Verification

OFFICIAL DOCUMENT

OFFICIAL INTERNSHIP AWARD RECORD

Earthix Software Solutions Pvt. Ltd.

This certificate is proudly presented to **Md Asrar** for successfully completing the **Full Stack Developer Internship** from 01 June 2026 to 31 July 2026 (8 Weeks).

Citation of Performance:
"The intern demonstrated excellent technical skills, dedication, problem-solving ability, and a strong willingness to learn while making valuable contributions to full stack projects including frontend, backend, API development, and database management."

EMP ID: EX321151

Cert ID: ESSPT321151

Issued: 01/08/2026



Verified Credential

Authorized by **Director Md Sanaullah**
Earthix Software Solutions Pvt. Ltd.
Official Portal: www.earthix.in

✓ **Authenticated Completion**



EARTHIX SOFTWARE SOLUTIONS PVT. LTD.

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CERTIFICATE OF COMPLETION

Full Stack Development Internship

Earthix Software Solutions Pvt. Ltd.

THIS CERTIFICATE IS PROUDLY PRESENTED TO

MD ASRAR

For Successfully Completing the

Full Stack Developer Internship

This certificate is awarded in recognition of the successful completion of the **Full Stack Developer Internship** at **Earthix Software Solutions Pvt. Ltd.**

The internship was successfully completed from **01 June 2026** to **31 July 2026** for a duration of **Two Months (8 Weeks)**.

During the internship period, the intern demonstrated excellent technical skills, dedication, problem-solving ability, and a strong willingness to learn while making valuable contributions to the full stack development projects including frontend, backend, API development and database management.

Employee ID : EX321151

Issued Date : 01/08/2026

Certificate ID : ESSPT321151

Earthix Software Solutions Pvt. Ltd.
Md Sanaulah
Director

Direct
Earthix Software Solutions Pvt. Ltd.

Earthix Software Solutions Pvt. Ltd.
Ph : 9217232660
www.earthix.in

Authorized
Earthix Software Solutions Pvt. Ltd.



Thank You!

Successfully concluded 8-Week Professional Full Stack Development Internship

Md Asrar

Employee ID: EX321151 | **Certificate ID:** ESSPT321151

IILM University, Greater Noida | Full Stack Developer Intern — Earthix Software Solutions

Contact: +91 8340452870 | Corporate Portal: www.earthix.in